Proposed Framework for Simultaneous Optimization of Evacuation Traffic Destination and Route Assignment

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Proposed Framework for Simultaneous Optimization of Evacuation Traffic Destination and Route Assignment

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In the conventional evacuation planning process, evacuees are assigned to fixed destinations mainly on the basis of geographical proximity. However, the use of such prespecified destinations (an origin–destination table) almost always results in less-than-optimal evacuation efficiency because of uncertain road conditions, including traffic congestion, road blockage, and other hazards associated with the emergency. By relaxing the constraint of assigning evacuees to prespecified destinations, a one-destination evacuation (ODE) concept has the potential to improve evacuation efficiency greatly. To this end, a framework for the simultaneous optimization of evacuation traffic distribution and assignment is proposed. The ODE concept can be used to obtain an optimal destination and route assignment by solving a one-destination (1*D*) traffic assignment problem on a modified network representation. When tested for a countywide special event–based evacuation case study, the proposed 1*D* model presents substantial improvement over the conventional multiple-destination (*nD*) model and can potentially reduce overall evacuation time by more than 60%. More important, this framework can be easily implemented and its efficiency enhanced simply by instructing evacuees to head for destinations resulting from the 1*D* simulations run beforehand.

As a result of heightened interest in emergency management and evacuation operations, transportation professionals have turned increasingly to evacuation modeling and planning. Most previous studies in evacuation modeling extended the applications of conventional transportation planning models, for example, the four-step process, to tackle the problem of evacuation. These planning models were designed for day-to-day travel under normal situations in which origins and destinations are easy to determine and, for practical purposes, remain static over time. But emergency evacuation calls for expeditiously mobilizing and transporting a sizable population out of harm's way and overcoming temporal and spatial constraints while at times facing uncertain road conditions due to congestion, road blockage, and other hazards associated with the emergency.

Although the transient and potentially chaotic nature of evacuation makes it far more challenging to manage than normal traffic operations, evacuation does have a unique advantage as far as planning goes. That is, although each motorist must head for and arrive at a specific destination under normal conditions, such motorists become more flexible in emergency evacuation conditions. In other words, as long as an evacuee safely exits the evacuation zone in a timely fashion, it is not so important which route is taken or at which location the evacuee leaves the zone. This flexibility in an evacuee's destination selection, and its associated benefit to the planning process, is neither well understood nor often exploited. In fact, this flexibility is often sacrificed through the common practice of rather rigidly assigning evacuees to designated routes, shelters, or destinations mainly on the basis of the simple criterion of proximity.

Past experience suggests that a major problem in evacuation operations is that routes exiting an evacuation zone are often limited in number and insufficient in capacity to handle the unusual surge in demand resulting from a large-scale emergency evacuation (*1, 2*). In most cases, constructing new routes and increasing roadway capacities are simply too cost-prohibitive to be considered practical for improving evacuation efficiency. Therefore, finding ways to improve the planning and operational aspects of the evacuation process to maximize the utility of the existing transportation network has often been the focus of past studies. Some such efforts have included the implementation of counterflow and contraflow lanes (*3*), staggering departure times (*4, 5*), traffic signal control, multijurisdiction coordination, special routing consideration for heavy vehicles (*6*), and so on. Yet no effort had recognized or explored the aforementioned flexibility in evacuee destination selection until Yuan and Han (*7*) proposed the concept of the most-desirable destination for evacuees.

In their study, Yuan and Han recognized that the distribution and assignment of evacuees to available routes and destinations has routinely been performed by municipalities with so-called local knowledge and past experience and that it has been practiced more as an art than as a science. The origin–destination (O-D) table, used in most evacuation models, therefore represents a fixed and static assignment that almost never leads to optimal efficiency as traffic demand and roadway conditions continue to fluctuate and shift throughout the evacuation period. Even with the implementation of dynamic traffic assignment (DTA), which is supposed to find the best routes for evacuees while taking into consideration changing traffic conditions, models based on fixed O-D tables can be quite inefficient when a destination becomes hard (or even impossible) to access because of congestion or blockage (*7*). In such situations, the flexibility of evacuees to head for alternative exit routes and destinations should be explored, which is the aim of this study.

Which comes first, destination assignment or route assignment for optimal evacuation? These two are really interrelated and constitute almost a chicken-and-egg situation. To optimize the routing

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problem, one has to know the destinations; yet to optimize the destination assignment, one has to know the minimal travel time, and hence route assignment, to all destinations. Some may consider that the complexity of this problem and the difficulty of its implementation render any potential gain in evacuation efficiency just an academic exercise. It will be demonstrated otherwise here. To tackle the inherent complexity of the problem, a framework is proposed for simultaneously optimizing evacuation traffic destination and route assignment. On the basis of this framework, the optimal evacuation destination and route assignment can be determined through a one-step optimization procedure. The formulation and potential applications of the proposed model are presented. Simulation analysis was conducted on a case study to test the model for a countywide special event evacuation. Practical implementation strategies are suggested and demonstrated as part of the case study.

METHODOLOGY

Given the spatial and temporal distributions of the evacuation demand for an evacuation area, or immediate response zone (IRZ), evacuation traffic destination distribution and route assignment can be optimized simultaneously on the basis of the concept of one-destination evacuation (ODE), in which the optimal destination assignment is determined along with the traffic assignment procedure on a modified network representation. Traditionally, an IRZ is the area closest to the site at which chemical munitions and agents are being stored until they can be destroyed. This zone, usually within a 6- to 9-mi radius of the stockpile, would require the quickest warning and response. People living or working in this zone may need to take protective measures quickly. The term IRZ is used here to represent the area in which evacuees should respond to an evacuation order immediately.

For a road network within an IRZ with *m* origins and *n* destinations, as shown in Figure 1*a*, the network can be modified and represented as the one in Figure 1*b*, where the original network is augmented with dummy links leading from each real-world destination point (road exiting the IRZ) to one common dummy destination point, denoted *D**. All these dummy links are assumed to have infinite capacity and zero cost (i.e., zero travel time). With this modification, a two-step decision-making process for the original evacuation network, including a demand distribution problem and a traffic assignment problem with *m* origins and *n* destinations (*m*-to-*n* assignment), is translated into a one-step decision-making process for the modified network, which is a traffic assignment problem with *m* origins and one destination (*m*-to-1 assignment). This *m*-to-1 traffic assignment problem can be solved by existing analytical algorithms or simulation methods toward user-optimal (UO) or system-optimal (SO) network flow patterns. On the basis of the traffic assignment results, the actual evacuation destination can be determined by tracing the real-world destination actually used by the evacuation traffic flowing from an origin to the dummy destination in the modified network. When the route selection and traffic assignment are optimized, the destination selection and demand distribution are optimized at the same time and provide a solution basis for evacuation planning and operations.

The proposed framework applies the ODE concept to reducing the number of flow conservation constraints in the traffic assignment problem and facilitating the search for a global optimum of the network flow patterns. The feasibility and superiority of this method are apparent when the mathematical formulation of the traffic assignment problem is examined. In the case of emergency evacuations, overall system performance is the major concern of the management, and thus the SO formulation and solution of the traffic assignment problem is of secondary importance. The SO traffic assignment problem, in its simplest form, is represented as follows (*8*):

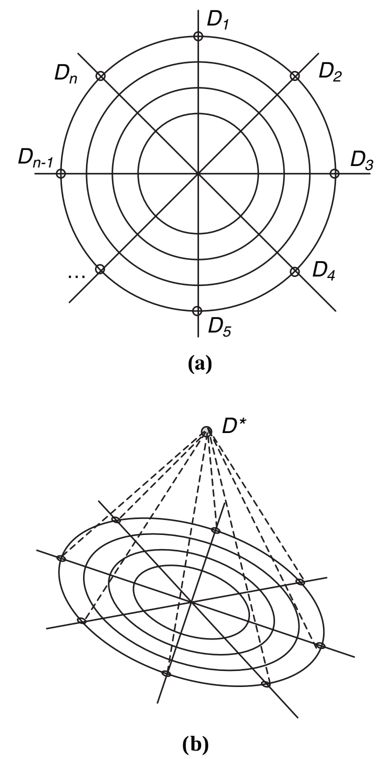

FIGURE 1 Transportation of destinations for IRZ network.

$$\min \sum_a f_a c_a(f_a) \tag{1}$$

subject to

$$f_a = \sum_{r \in R} \delta_{ar} h_r \quad \forall\, a \in A \tag{2}$$

$$\sum_{r \in R_{ij}} h_r = D_{ij} \quad \forall\, i \in I,\ j \in J \tag{3}$$

$$h_r \geq 0 \quad \forall\, r \in R \tag{4}$$

where

f_a = flow on link *a*,
$c_a(x)$ = cost function on link *a* with flow of *x*,
h_r = flow on path *r*,
D_{ij} = total demand between O-D pair *i* and *j*,
A = set of links,
R = set of all paths between all zone pairs,
R_{ij} = set of all paths from origin *i* to destination *j*,
I = set of origins,
J = set of destinations, and
δ_{ar} = link-path incidence variable; $\delta_{ar} = 1$ if link *a* is part of path *r* and $\delta_{ar} = 0$ otherwise.

The objective function, Equation 1, minimizes the total travel cost for all travelers in the network. The decision variables in this optimization problem are the flow levels on each link. The link flow levels that minimize the objective function must meet the following constraints: first, the total flow on a link must equal the summed flow of all paths that use that link (Equation 2). Second, the flow on all paths between an O-D pair must add up to the aggregate travel demand for that pair (Equation 3). Finally, all path flows must be nonnegative (Equation 4). When this formulation is adapted to a 1*D* traffic assignment problem, the objective function and most constraints remain the same except that the set of flow conservation constraints, Equation 3, is reduced and transformed to the following:

$$\sum_{r \in R_i} h_r = D_i \qquad \forall\, i \in I \tag{5}$$

Obviously, a solution of the link flows $\{f_a\}$ to the traffic assignment problem on a network with multiple destinations that satisfies all constraints (Equations 2, 3, and 4) and minimizes the objective function is also a feasible solution to the traffic assignment problem with the same objective function but with relaxed constraints (Equations 2, 4, and 5) on a modified network representation with one destination. However, the solution to the multiple-destination (*nD*) assignment problem is varied by the form of the demand distribution (*m*-to-*n* O-D table), whereas the solution to the one-destination (1*D*) traffic assignment problem is unique and independent from the demand distribution (*m*-to-1 O-D table). The minimum value of the objective function of the *nD* traffic assignment problem can only be equal to or larger than that of the objective function of the 1*D* traffic assignment problem. Theoretically, solving the *nD* traffic assignment problem based on a prespecified O-D table will only lead to a local minimum of the broader 1*D* traffic assignment problem. Therefore, the most optimal network flow patterns (destination and route assignments) can be solved with the 1*D* traffic assignment formulation.

Regardless of the formulation of the traffic assignment problem (i.e., static or dynamic, UO or SO), flow-conservation constraints must be satisfied. Hence, the transformation discussed earlier can be equally applied. Reducing the number of the flow-conservation constraints will not change such mathematical properties as the existence, uniqueness, and stability of the solution to the original formulation, but this approach would facilitate reaching a more- or most-optimal solution. In fact, 1*D* trip distribution setup makes it easier to formulate the DTA problem.

The physical behavior of traffic on a roadway link generally exhibits the so-called first-in, first-out (FIFO) property; that is, traffic that enters a link at a particular time exits the link, on average, before traffic that enters at a later time. The FIFO requirement creates an inherent difficulty in all mathematical programming approaches to the time-dependent assignment problem. The *nD* assignment problem requires additional constraints to explicitly satisfy the FIFO requirement, but these constraints will make the feasible region nonconvex, destroying many of the computational and mathematical advantages of the formulation. However, the FIFO requirement is easily satisfied in the 1*D* formulation (*9*).

POTENTIAL APPLICATIONS

The proposed framework for simultaneously optimizing evacuation traffic destination distribution and route assignment is a theoretical model deployable to real-world problems. Not only does the resulting simpler 1*D* assignment problem have an exact solution through analytical methods (*10, 11*), but the concept of ODE can also be applied to emergency evacuation planning through the use of simulation software that has the capability of static or dynamic traffic assignment.

Simulation-based assignment models, especially at the microscopic level, have the advantage of modeling complex network conditions and multiple management strategies that may improve the efficiency of evacuation operations but are generally too cumbersome to formulate by analytical methods. These models include intersection control, contraflow operations, and certain aspects of intelligent transportation systems (ITS) (*3–6*). On the basis of the concept of ODE, existing microscopic simulation models can be used without modification for evacuation modeling and optimization. In fact, various operational strategies can still be considered under ODE. Ultimately, the ODE model can be implemented in real time, taking advantage of the ITS infrastructure in the IRZ. Even at the planning level, ODE provides a tool for assessing and greatly enhancing existing evacuation plans.

To demonstrate the feasibility and benefits of the proposed ODE model for improving evacuation efficiency and also to demonstrate its applicability, several case studies of real-world evacuation operations at different locales were conducted with different simulation software packages. One of these cases is presented next.

CASE STUDY

A countywide special event evacuation operation was simulated with DYNASMART_P (*12*) for Knox County, Tennessee, which has a population of around 382,000 (or 158,000 households) (*13*). Two major Interstates, I-40 and I-75, cross the study area. Evacuation of the entire Knox County was modeled. Two network setups or modeling strategies, *nD* and 1*D*, were simulated and compared for their effectiveness.

Simulation Tool

DYNASMART_P is a dynamic traffic operational planning tool developed under the FHWA DTA research initiative. It integrates traffic flow models, path-processing methodologies, behavioral rules, and information supply strategies into a single simulation-assignment framework. The model is designed to assign time-varying traffic demands and to model the corresponding traffic patterns in order to evaluate overall ITS network performance. DYNASMART-P is able to model the evolution of traffic flows in a traffic network that result from the decisions of individual travelers seeking the best paths en route over a given planning horizon (*12, 14*).

Case-Site Modeling

The Knox County roadway network (Figure 2) was geocoded for DYNASMART_P with detailed geographic representation including on- and off-ramps and interchanges as well as traffic control device information. It consists of 263 zones, 1,006 links, and 2,179 nodes, of which 120 are signalized (see Figure 3).

The transportation analysis zones (TAZs) defined for the study area generally follow those formulated by the Knoxville and Knox County Metropolitan Planning Commission. Evacuation demand was estimated on the basis of the number of households in each TAZ. According to U.S. census data for 2000, the number of households was aggregated at the TAZ level and is shown in Figure 3. In total, 157,733 vehicle trips were estimated and modeled for the study area for a hypothetical countywide evacuation event.

FIGURE 2 Geocoded highway network for Knox County.

Evacuation trips are generally directed outward as evacuees leave the IRZ and seek safety. There is a total of 24 highway exits for the IRZ. With the conventional approach, individual destination zones are defined at the end of each network exit and thus form an *nD* evacuation network. For this *nD* network, two strategies—nearest-exit (proximity) assignment and Interstate-biased assignment—were implemented (see Figures 4 and 5, respectively). All evacuees from the same TAZ are assumed to evacuate toward the same exit zone in adherence with the assignment strategies. The performance of the *nD* model based on these two different exit strategies is to be compared with that of the 1*D* model, which is constructed by connecting the 24 exit points in the real world to one dummy destination that all evacuees head for via whatever routes.

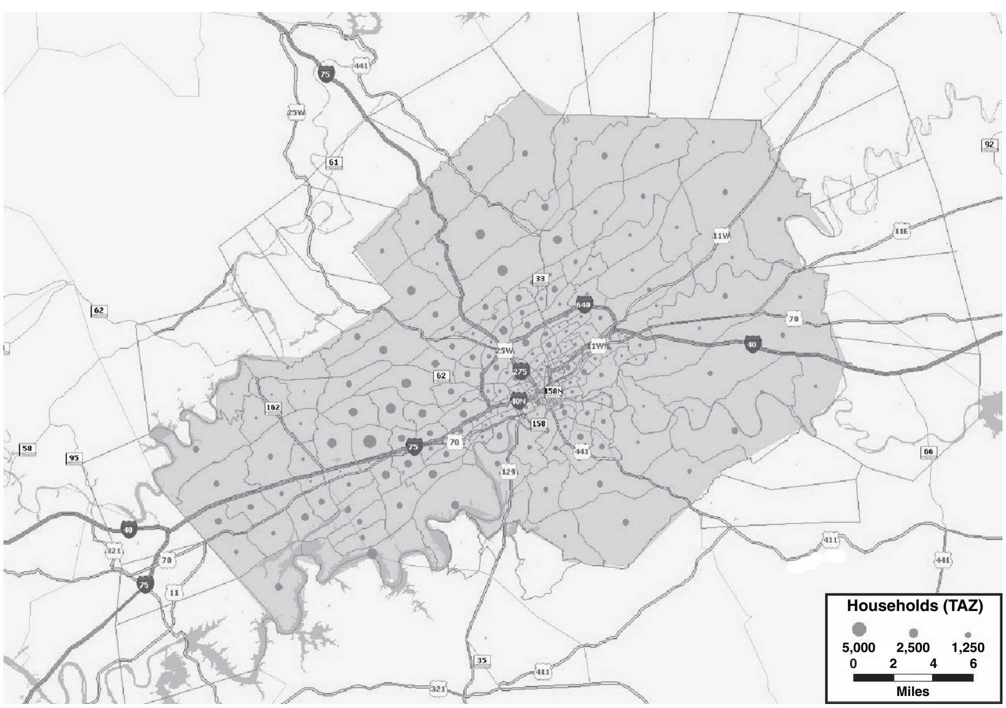

FIGURE 3 Number of households by TAZ in Knox County.

FIGURE 4 Nearest-exit–based destination assignment.

Scenarios and Simulation Results

Eight scenarios representing various destination and route assignment strategies were investigated with DYNASMART_P. The evacuation traffic was loaded onto the network in a uniform fashion during the first hour of the simulation. Because of the hefty computational effort required to simulate hundreds of thousands of vehicles in a large network, all evacuation durations were capped at 20 h.

All eight scenarios and their results are summarized in Table 1, in which the times required to evacuate 25%, 50%, 75%, and 95% of the evacuees are shown together with the time required to evacuate 100% of the population, a target that is seldom achieved in reality but is provided nevertheless for reference purposes.

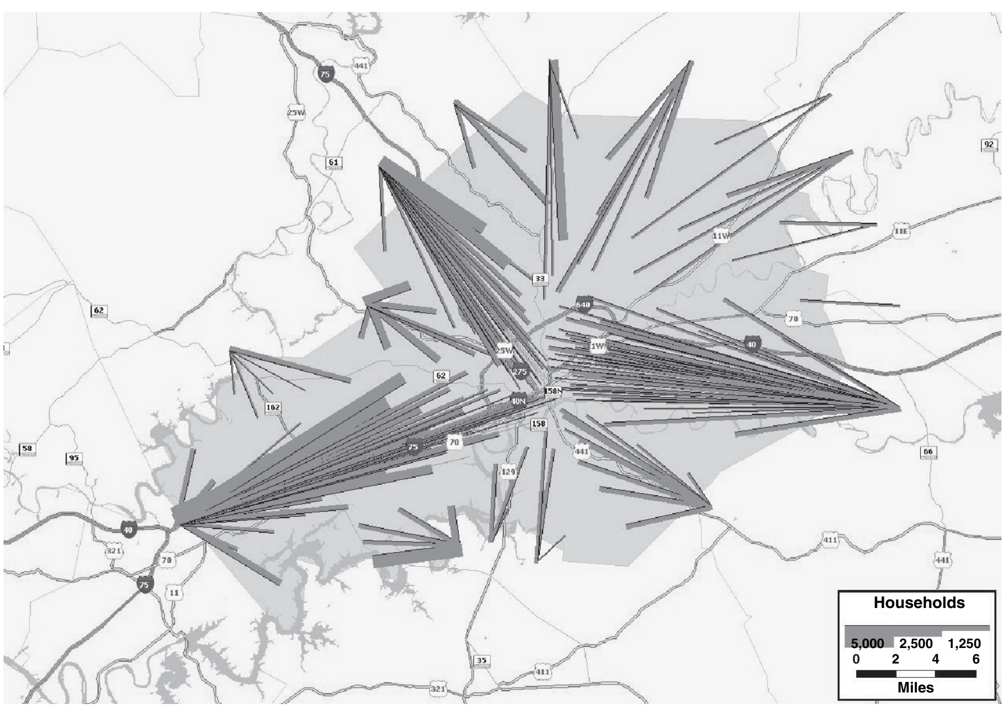

FIGURE 5 Interstate-biased destination assignment.

TABLE 1 Evacuation Times of All Assignment Scenarios

		Percent Evacuated				
		25%	50%	75%	95%	100%
Scenario		Evacuation Time in Minutes				
SR						
1	Nearest exit[1,5]	115	305	890	—	—
2	Interstate-biased[2,5]	105	260	755	—	—
3	1*D* SR	65	125	235	490	1040
DR						
4	Nearest exit[3]	115	300	835	—	—
5	Interstate-biased[4]	105	250	480	1065	—
6	1*D* DR	60	120	180	225	345
Improved SR						
7	*nD* w/ 1*D* SR O-D[5]	70	160	280	820	1120
8	*nD* w/ 1*D* DR O-D[5]	60	130	220	460	970

SR = static routing; DR = dynamic routing.
[1]Only 82.1% of the population was evacuated in 20 hours.
[2]Only 81.7% of the population was evacuated in 20 hours.
[3]Only 83.7% of the population was evacuated in 20 hours.
[4]Only 99.1% of the population was evacuated in 20 hours.
[5]Scenario can be implemented without real-time traffic information.

Figure 6 compares the results from the first, second, seventh, and eighth scenarios, four approaches that can actually be implemented without real-time en route information. Without any capital investment in ITS infrastructure, contraflow control devices, and so forth, and simply by instructing evacuees to head for different destinations beforehand, time savings in the range of 70% can be achieved.

Nearest-Exit Static Routing

Nearest-exit static routing (SR) is the base case, in which evacuees are assigned to the nearest of the 24 exits (Figure 4). The conventional *nD* assignment model is used here with SR, which does not provide real-time traffic information or allow en route dynamic routing (DR). This base case is the typical evacuation planning strategy. The simulation results indicate a long evacuation process that would leave about 18%, or close to 70,000 people, stranded in the IRZ 20 h after the evacuation operation commenced. The evacuation curve is shown as a dashed line in Figure 7.

Interstate-Biased SR

Interstate highways are perceived to have desirable access control, faster operational speed, higher roadway capacity, and superior geometric design and to be more dependable in time of emergency. By assigning most evacuees to the nearest of the four Interstate highway exits (see Figure 5), it was hoped that some level of improvement in evacuation efficiency might be achieved. This scenario is still an *nD* assignment. The simulation results suggest a slightly better evacuation curve at first (see solid line in Figure 7); however, this scenario did not outperform the nearest-exit SR scenario at the end of the 20-h period, at which point around 18% of the population was still stranded in the IRZ.

1*D* SR

With the concept of ODE assignment, every evacuee was assigned to the same dummy destination to which all 24 exit points led. Each trip, when loaded onto the network, was assigned to the best route at the time. However, as in all SR assignments, vehicles once en route are not afforded real-time traffic information for rerouting, which in practical terms fixed the real-world destination for each vehicle. The result, which is shown as a dotted line in Figure 7, suggested a remarkable improvement in evacuation efficiency over the conventional *nD* approaches, represented by the first two scenarios, which can barely evacuate 80% of the population in 20 h; in contrast, the 1*D* SR moves 95% of the population, the typical measure of effectiveness for evacuation operations, in about 8 h. In general, the 1*D* SR offers a time savings of 70% or so in evacuating the IRZ population. One caveat,

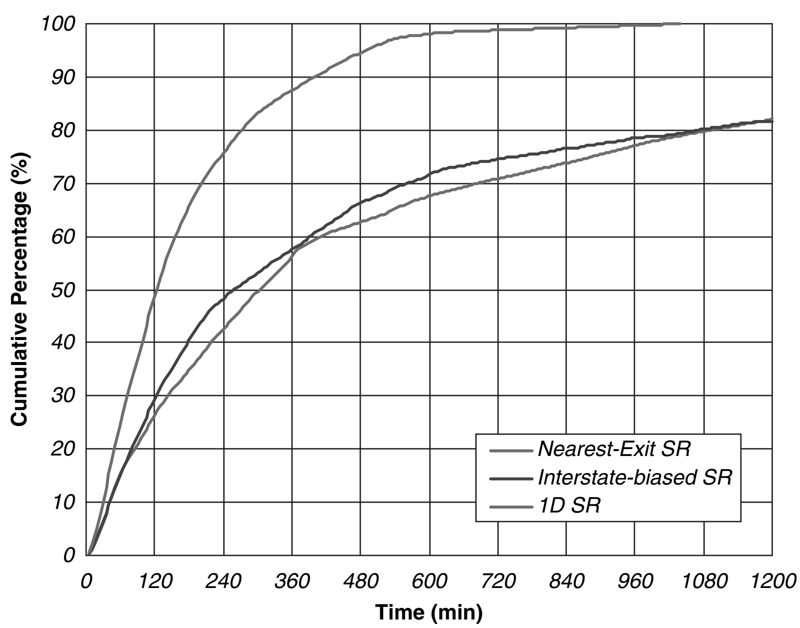

FIGURE 6 Evacuation curves based on improved O-D tables with SR: percentage of evacuees cleared from IRZ.

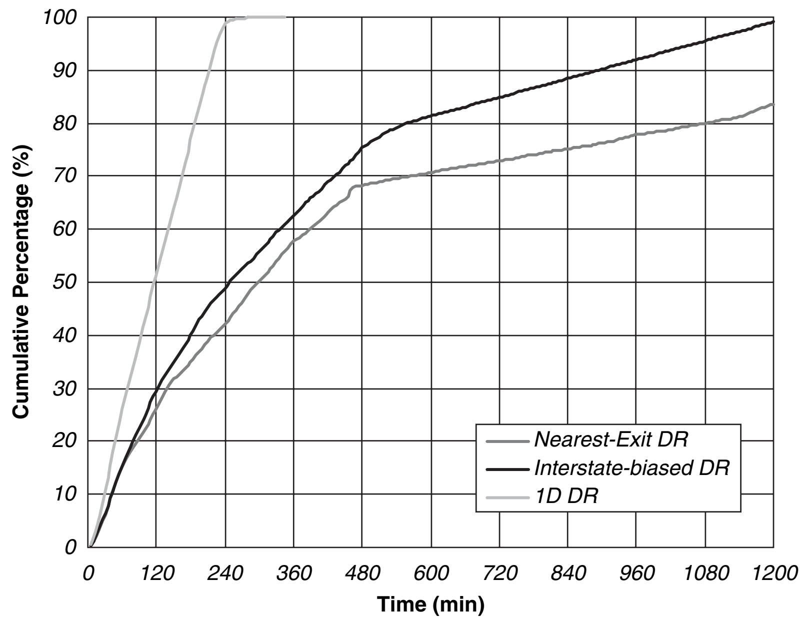

FIGURE 7 Evacuation curves with SR: percent of evacuees cleared from IRZ.

though, is that because the evacuees were, in effect, assigned to the best destination at the time of departure, this scenario requires real-time information and hence makes it difficult to implement in areas without a deployed or matured ITS infrastructure.

Nearest-Exit DR

This scenario is similar to the first one, but vehicles are now being dynamically routed on the basis of real-time traffic information, a function of DYNASMART_P that emulates the effects of the advanced traveler information system (ATIS) (*10*). Each evacuee still heads toward a preassigned destination but perhaps via a better route. This operation is the typical DR applied to the conventional *nD* evacuation plan. Obviously, some level of ITS infrastructure has to be deployed for this scenario to work in the real world.

The results are shown in Table 1 and as the dashed line in Figure 8. It is somewhat surprising that DR did not seem to improve the evacuation time in a noticeable fashion. At the end of the 20-h evacuation period, more than 60,000 people were still stranded in the IRZ. This

FIGURE 8 Evacuation curves with en route DR: percent of evacuees cleared from IRZ.

finding is perhaps an indication of the weakness of the conventional proximity-based destination assignment approach, which would not gain much efficiency even if it were able to take advantage of expensive ITS infrastructures and sophisticated routing schemes because of the less-than-optimal destination assignment at the outset.

Interstate-Biased DR

This scenario is similar to the second one, but DR is applied to improve routing in real time. This is still a conventional *nD* assignment. But with real-time rerouting, efficiency should improve. The results of this scenario, shown as the solid line in Figure 8, are more promising, since 95% of the population was evacuated in about 16 h. About 3,000 people were still stranded in the IRZ at the end of the 20-h evacuation period, but this result is already a significant reduction from the 60,000 in the nearest-exit DR scenario. It appears that the DR approach is more effective when most evacuees were destined for Interstate exits. But once again, it should be kept in mind that the implementation of the DR scenario would require real-time en route information.

*1*D *DR*

This scenario is similar to 1*D* SR, but in addition to ODE, DR is applied. Intuitively, this scenario seems the most efficient of all strategies because it simultaneously optimizes the destination and route assignment on the basis of real-time information for all evacuees. The results, shown as the dotted line in Figure 8, certainly support this notion. The large population, which could not be cleared in 20 h in four of the first five scenarios even though DR was also deployed in two of them, was easily evacuated in just over 4 h, an astonishing time savings of more than 80%. Although the implementation of DR requires ITS and hence makes this optimal strategy rather challenging to realize, the results could still provide valuable insights for improving non-DR strategies. The short evacuation time serves as a lower bound against which other scenarios can be measured and that they should strive to achieve. More important, the resultant O-D table can be used for implementation with the *nD* SR approach.

*1*D *SR–Optimized* nD *SR*

Ultimately, these destination and routing assignments are worthwhile only if they can be applied to real-world emergency situations. Although the concept of ODE may be superior on paper, the benefit to the public is minimal if it cannot be implemented. Constraints in the real world include the following: (*a*) evacuees would not appreciate being told to head for a dummy destination in a time of emergency and (*b*) ITS infrastructure is not readily deployed in all locales and even when available may not always be reliable under severe weather and other emergency conditions. To address the first issue, the resultant O-D table from the 1*D* SR simulations was adopted to construct a new *nD*-based O-D table, as shown in Figure 9, which should outperform the first two *nD* scenarios. The results, shown as the dotted line in Figure 6 and also in Table 1, are reassuring. Without DR or real-time information, this *nD* traditional assignment was able to clear 95% of the population within 14 h and, in general, exhibited a time savings of better than 60% in comparison with the basic scenario.

*1*D *DR–Optimized* nD *SR*

Similar to the concept in the previous scenario, the resultant O-D table from the 1*D* DR is now used to construct a new *nD*-based O-D table, as shown in Figure 10. Since the 1*D* DR is the most efficient approach, simultaneously optimizing both destination and routing problems, it was expected that the resultant O-D table would also yield efficient *nD* SR assignment, an expectation that the simulation results, shown as a solid line with X's in Figure 6, clearly support. Again, without using DR or real-time information, this *nD* traditional assignment was able to clear 95% of the population

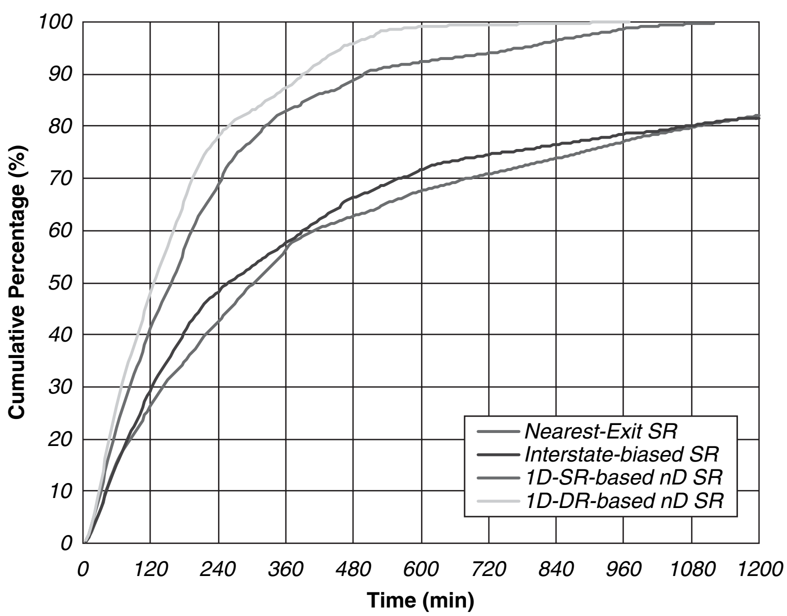

FIGURE 9 Destination assignment from 1*D* optimization with SR.

FIGURE 10 Destination assignment from 1*D* optimization with DR.

within 8 h and, in general, exhibited a time savings of better than 70% in comparison with the basic scenario.

CONCLUSIONS

In this study, a framework for the simultaneous optimization of evacuation traffic distribution and assignment was proposed. Based on the concept of ODE, the optimal destination and route assignment can be obtained together by solving a 1*D* traffic assignment problem on a modified network representation. For a hypothetical countywide evacuation case study, the proposed 1*D* method presents a substantial improvement over the conventional *nD* model for both planning and operational purposes: for instance, a nearly 80% reduction in the overall evacuation time can be achieved when traffic routing is modeled with en route information in the 1*D* framework, and the 1*D* optimization results can also be used to improve the planning O-D tables, resulting in 60% to 70% reduction in the overall evacuation time.

All these results demonstrate that 1*D* modeling is an efficient method of optimizing network flow patterns and reducing evacuation time by providing flexibility in destination selection and consequently route selection. More important, this framework can be easily implemented and the time savings realized just by instructing evacuees to head for destinations derived from the 1*D* model beforehand.

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